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PS – 391

II Semester M.C.A. Examination, Nov./Dec. 2022
(CBCS) (2020-21 and Onwards)
COMPUTER SCIENCE
2MCA6 : Artificial Intelligence

Time : 3 Hours Max. Marks : 70

***Instruction** : Answer **all** the Parts.*

PART – A

Answer **any five** of the following. **Each** question carries **six** marks : **(5×6=30)**

1. What is AI ? Explain its applications in detail.
2. What is an intelligent agent ? Explain its structure with a neat diagram.
3. Explain reasoning in proposition logic using Wumpus world example.
4. What is planning ? Explain components of planning.
5. What is Natural language processing ? Explain all its components in brief.
6. Define expert system and explain its architecture with a neat diagram.
7. What are neural networks ? Explain briefly working of ANN.
8. What is learning ? Explain inductive learner with an example.

PART – B

Answer **any four** of the following. **Each** question carries **ten** marks : **(4×10=40)**

9. Prove that "Robert is criminal" using forward chaining using the following statements.
"As per the law it is a crime for an American to sell weapon to hostile nation. Country A, an enemy of America has some missile and all missiles were sold to it by Robert, who is an American citizen." **10**

10. a) What is uncertainty ? Explain probabilistic reasoning in AI. **5**

    b) Explain fuzzy logic ? **5**

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11. a) What are robotics ? Explain the components of a Robot. 5

    b) Explain various parsing techniques. 5

12. a) Briefly explain difference between machine learning and deep learning. 5

    b) Discuss the nature of environment. 5

13. Write short note on :

    a) Constraint satisfaction problem. 5

    b) Knowledge engineering in FOPL. 5

14. What is a problem ? Explain how a problem can be represented in state space search using vaccum world example. 10

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II Semester M.C.A. Examination, Nov./Dec. 2023

(CBCS) (2020-21 and Onwards)

COMPUTER SCIENCE

2MCA6 : Artificial Intelligence

Time : 3 Hours Max. Marks : 70

Instructions : *1) Answer **any 5** from Section – **A**.*

*2) Answer **any 4** from Section – **B**.*

SECTION – A

Answer **any 5** questions from the following. **Each** question carries **6** marks :

1. a) Define agent. Mention types of agent in AI. 3

   b) State the relationship between agent and environment. 3

2. Prove that uniform-cost search and breadth-first search with constant step cost are optimal when used with the graph-search algorithm. Show a state space with constant step costs in which graph search using iterative deepening, find a sub optimal solution. 6

3. Discuss 8-queens problem. How Euclidean distance and Manhattan distance are calculated ? Give any one application of each. 6

4. What is learning ? Explain various learning techniques. 6

5. Draw a semantic network representing the following knowledge. 6

   a) Every vehicle is a physical object.

   b) Every car is a vehicle.

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c) Every car has 4 wheels.

d) Electrical system is a part of car.

e) Battery is a part of electrical system.

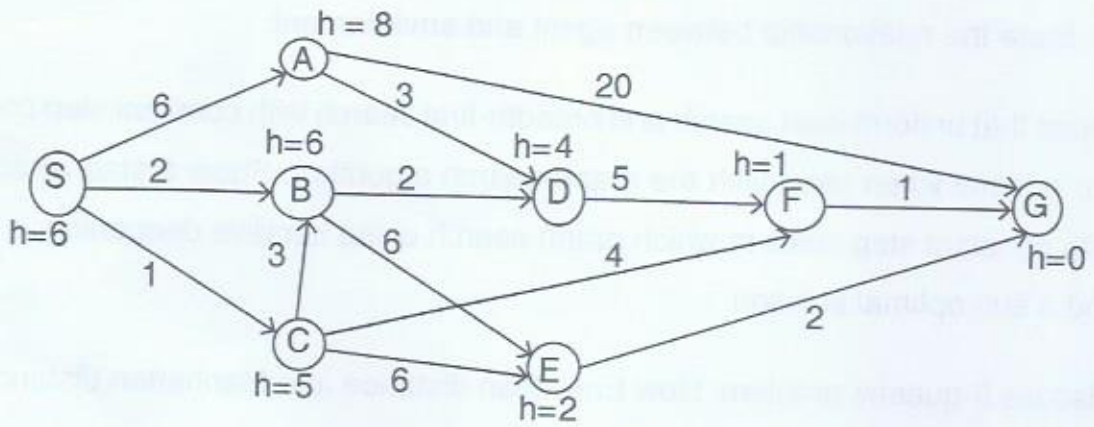
f) Pollution system is a part of every vehicle.

g) Vehicle is used in transportation.

h) Swift is a car.

6. Define planning. Explain the needs of planning. 6

7. Consider the search problem below with start state S and goal state G. The transition costs are next to the edges and heuristic values are next to the state. What is the final cost using A* search ? 6



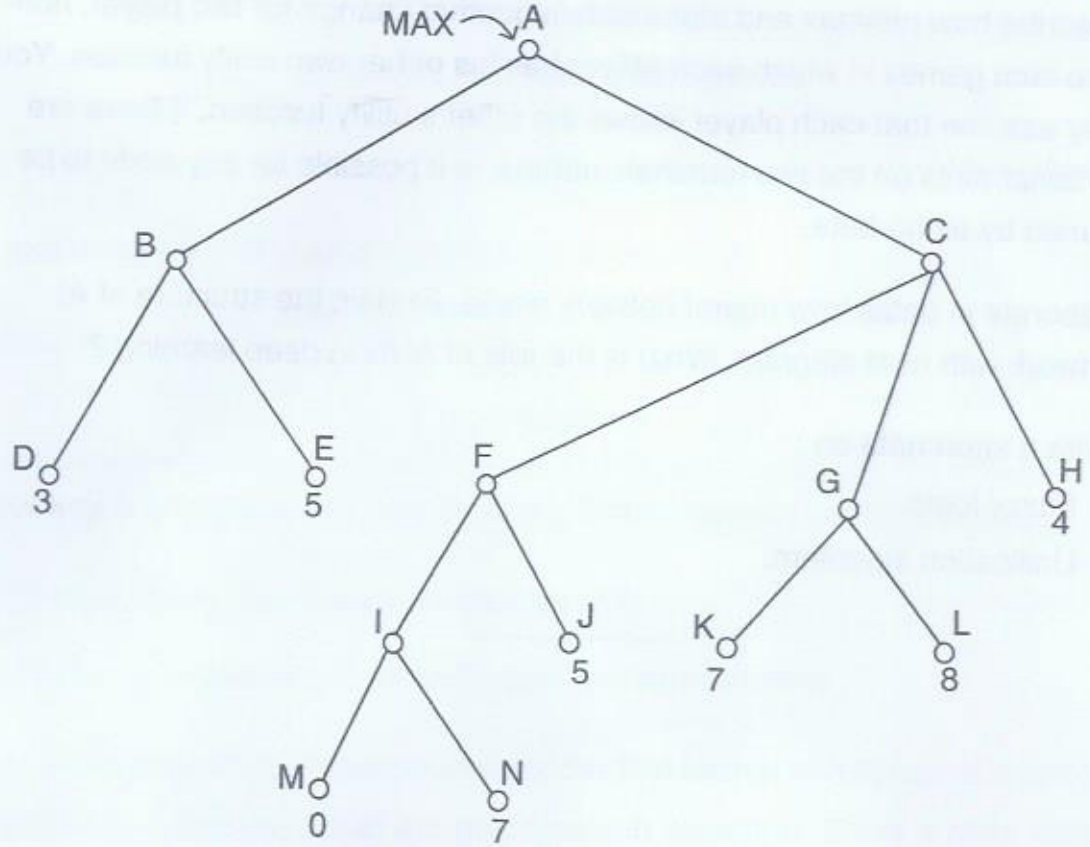
8. a) What is natural language processing ? Explain different stages of NLP and its advantages. 3

b) How machine learning is different from deep learning ? Explain any two system learning. 3

SECTION – B

Answer **any 4** questions. **Each** question carries **10** marks :

9. Consider the following game tree. **10**



a) Find the best move for the MAX player using the minmax procedure.

b) Perform a left-to-right alpha-beta pruning on tree. Indicate where the cut offs occur.

10. Consider the following sentences. **10**

a) John likes all kinds of food.

b) Apples are food.

c) Chicken is food.

d) Anything anyone eats and is not killed by food.

e) Bill eats peanuts and is still alive.

f) Sue eats everything Bill eats.

Translate these sentences in to formulas in predicate logic. Prove that "John likes peanuts" using background chaining. Convert the formula into clause form.

11. Explain the architecture of expert system. What are the advantages and limitations of expert system ? **10**

12. Describe how minmax and alpha-beta algorithm change for two player, non-zero sum games in which each player has his or her own utility function. You may assume that each player knows the other's utility function. If there are no constraints on the two terminals utilities, is it possible for any node to be pruned by alpha-beta. **10**

13. Elaborate in detail how neural network works. Explain the structure of AI network with neat diagram. What is the role of ANN in deep learning ? **10**

14. Write a short note on :
    a) Fuzzy logic. **5**
    b) Unification algorithm. **5**

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II Semester M.C.A. Examination, December 2024

(CBCS) (2020-21 and Onwards)

COMPUTER SCIENCE

2MCA6 : Artificial Intelligence

Time : 3 Hours Max. Marks : 70

Instruction *: Answer* ***any five*** *questions from Section –* ***A*** *and* ***any four*** *from Section –* ***B****.*

PART – A

Answer **any five** questions. **Each** question carries **6** marks. **(5×6=30)**

1. What is an agent ? Explain its structure and environment.
2. Solve the Water Jug problem : You are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug ? Explicit assumptions : A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available. Explain it as a state space search.
3. Write A* algorithm and discuss briefly the various observations about algorithm.
4. What is knowledge ? And different types of knowledge. Discuss about declarative and procedural knowledge.
5. Write a note on Truth Maintenance System.
6. Explain the role of probabilistic reasoning in reasoning with uncertainty.
7. What is Parse Tree and its role in NLP and discuss with an example.
8. What are the properties of Neural Networks ? List out the problem areas addressed by Neural Networks.

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PART – B

Answer **any four** questions. **Each** question carries **10** marks. **(4×10=40)**

9. What is "Artificial Intelligence and Artificial Intelligence Technique" ? Briefly explain how AI technique can be represented and list out some of the task domain of AI.

10. Consider the following sentences :

    - John likes all kinds of food
    - Apples are food
    - Chicken is food
    - Anything anyone eats and isn't killed by is food
    - Bill eats peanuts and is still alive
    - Sue eats everything bill eats.

    i) Translate these sentences into formulas in predicate logic

    ii) Prove that John likes peanuts using backward chaining

    iii) Convert the formulas of a part into clause form

    iv) Prove that John likes peanuts using resolution.

11. List out the planning terminologies and components of planning for the below task.

    "Suppose we have a fixed supply of paint; some white, some pink and some red. We want to paint a room so that it has light red walls and a white ceiling. We could produce light red paint by adding some white paint to red. But then we could not paint the ceiling white. So this approach should be abandoned in favor of mixing the pink and red paints together".

12. What is game playing ? And discuss about Alpha-beta pruning strategy with an example.

13. What is learning and different types of learning, discuss about explanation based learning with an example.

14. Define expert system. Explain the architecture of an expert system in detail with a neat diagram and an example.

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II Semester M.C.A. Examination, November/December 2025

PG – 1431

(CBCS Scheme) (2020 – 21 and Onwards)

COMPUTER SCIENCE

2MCA6 : Artificial Intelligence

Time : 3 Hours

Max. Marks : 70

Instruction : *Answer **any five** questions from Part – **A** and **four** questions from Part – **B**.*

PART – A

I. Answer **any five** questions : **(5×6=30)**

1) Explain the difference between Natural Intelligence and Artificial Intelligence.

2) Consider problem of Magic square where "a square of same number of rows and columns, where consecutive numbers are filled in blank spaces of each row and column such that numbers in each row, column and two diagonals, add up to the same number. The numbers have to be consecutive and one number can be used only once." Write the algorithm to solve it and also devise the production rules and its state space.

3) What is Alpha-Beta pruning ? On what factor the efficiency of the Alpha-Beta procedure depends ?

4) Consider the following set of axioms

i) Everyone who loves someone who loves them back is happy.

ii) Mary loves everyone.

iii) There is someone who loves Mary.

From the above statements conclude that : Mary is happy.

5) Given the following initial and the goal state for the Blocks world problem, construct a set of operators (Rules) and hence generate a plan to reach the goal state from the initial state.

Initial state: ontable(A) $\wedge$ ontable(B) $\wedge$ on(C, B) $\wedge$ clear(A)

Goal State : ontable(B) $\wedge$ on(C, B) $\wedge$ on(A, C) $\wedge$ clear(A)

Where ontable(x) : block x is on top of the table on(x, y): block x is on top of block y clear(x) : there is nothing on top of block x; therefore it can be picked up handempty: you are not holding any block.

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6) Define Natural Language Processing. Discuss the syntactic processing of the NLP with an example.

7) Why planning is considered as an important component of intelligence ? Give example to support the answer.

8) What is an expert system ? Briefly describe five major components of an expert system.

PART – B

II. Answer **any four** questions : **(4×10=40)**

9) a) Differentiate between blind search and heuristic search techniques. When is blind search better than heuristic search ? **5**

b) Assume one has to move from place "A" in the city of Bengaluru, to place "B" in the city of Delhi. The available means of transport are walking, self driving, taking a bus, taking a cab, taking a train or flying. **5**

Discuss the means end analysis for the above situation, how can we achieve the operator subgoaling ?

10) What are knowledge and knowledge engineering and their characteristics ? Discuss the various types of knowledge. Explain them giving example for each. **10**

11) What is learning ? Discuss different types of learning. Explain various approaches in explanation based learning. **10**

12) Generate the Parse tree for the following sentences **(5×2=10)**

a) Piyush wanted to go to the market with Adarsh.

b) I heard the story listening to the radio.

c) I heard the kids listening to the radio.

d) All books and magazines that deal with controversial topics have been removed from the syllabus.

e) India will be superpower by the year 2040.

13) What is semantic net representation of knowledge ? Construct a semantic net representation for the following : **(2×5=10)**

a) Every batter hit a ball.

b) All the batters like the pitches.

14) Write short notes on **(2×5=10)**

a) Waltz algorithm.

b) RTN.

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